

Unlocking Societal & Administrative Insights from Aadhaar Enrolment Data

Transforming Digital Identity Logs into Actionable Governance Indicators

A Project for the UIDAI Data Hackathon 2026



From Data to Governance Action

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PROBLEM STATEMENT

Aadhaar enrolment data captures large-scale demographic and geographic movements of India's population. While this data is routinely collected, its potential for **anticipatory governance** remains underutilised.

Government agencies face persistent challenges such as:

- Sudden migration-driven pressure on enrolment infrastructure
- Uneven access to Aadhaar services across regions
- Planning gaps for welfare portability and service delivery

Currently, decision-making relies largely on **reactive measures** rather than early indicators.

The Challenge



UIDAI manages over 1.3 billion identity profiles—a massive asset currently treated as an operational log. This data is under-utilized for revealing societal patterns like migration shifts, service volatility, and exclusion risks.

The Solution



We propose three proprietary governance indices:

1. **AEVI**: Aadhaar Enrolment Volatility Index (Operational Efficiency)
2. **MPI**: Migration Pressure Indicator (Workforce Mobility)
3. **CESI**: Child Enrolment Stability Index (Social Protection)

The Outcome



Shifting from reactive administration to predictive governance.

- Optimization of Seva Kendra infrastructure.
- Evidence-based migrant welfare planning.
- Closing the legal identity gap for children.

Objective

To identify **meaningful patterns, trends, and volatility signals** within Aadhaar enrolment data that can:

- Support proactive infrastructure planning
- Indicate migration and access stress
- Enable data-driven governance without compromising privacy

DATASET USED

Dataset: Aadhaar Enrolment Dataset (UIDAI)

Nature: Aggregated and anonymised

Key attributes:

- **Dataset:** Aadhaar Enrolment Dataset
- **Source:** UIDAI (event.data.gov.in)
- **Records:** 1,006,029 aggregated entries
- **Time Range:** Up to December 2025
- **Granularity:** State, District, PIN code, Date
- **Age Categories:**
 - 0–5 years
 - 5–17 years
 - 18+ years

The dataset contains over **1 million records** spanning multiple states and time periods. No individual-level or biometric data is used.

METHODOLOGY

Data Cleaning & Standardisation

- Standardised inconsistent state names
- Converted dates into year and month
- Created total enrolment counts
- Removed non-informative or invalid entries

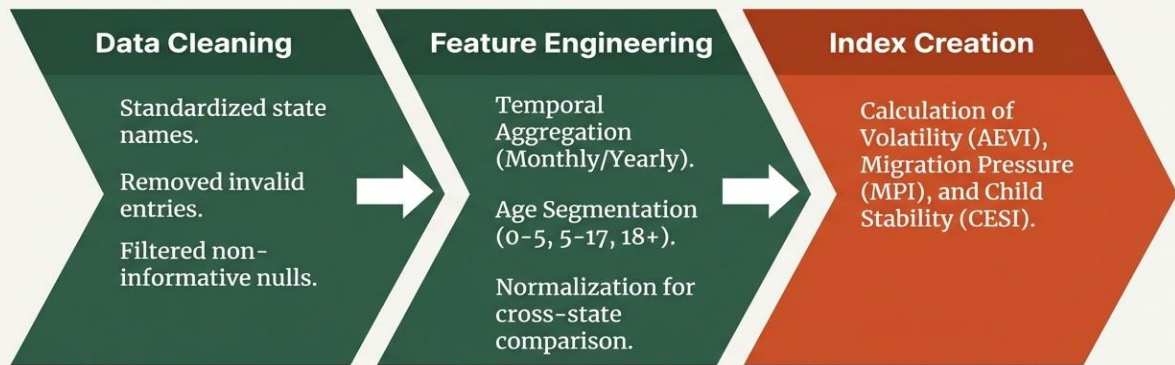
Feature Engineering

- Monthly and yearly enrolment aggregation
- Age-group-wise enrolment proportions
- Normalisation for cross-state comparison

Exploratory Analysis

- Temporal enrolment trends
- State-wise and age-wise distribution
- Volatility and concentration patterns

Methodology: Engineering Signal from Noise



Analytical Pipeline

Raw CSV Files



Data Cleaning & Standardisation



Date Parsing & Feature Engineering



Aggregation (Monthly, State-wise)



Indicator Construction



Visual & Statistical Analysis

Key Processing Steps

- State name normalisation (case & spelling unification)
- Date conversion to Year, Month, Month Name
- Creation of Total Enrolments feature
- Monthly aggregation for temporal stability
- Outlier-resilient normalisation

DATA ANALYSIS

Univariate Analysis

- Distribution of enrolments across age groups
- Identification of high-volume states

Bivariate Analysis

- Relationship between time and enrolment volumes
- State vs age-group dependency patterns

Trivariate Analysis

- State × Month × Age-group dynamics
- Detection of seasonal and migration-linked trends

Key Observation:

States with high adult enrolment volatility also exhibit pressure on enrolment infrastructure.

LOGIC BEHIND THE CODE

AEVI — Aadhaar Enrolment Volatility Index

Purpose: Measure enrolment instability over time

Policy use: Staffing optimisation and resource allocation

High AEVI values indicate irregular enrolment patterns due to migration, awareness drives, or operational inefficiencies.

$$\text{AEVI} = \frac{\text{Standard Deviation of Monthly Enrolments}}{\text{Mean Monthly Enrolments}}$$

MPI — Migration Pressure Indicator

Purpose: Detect migration-linked enrolment surges

Policy use: Infrastructure planning and temporary service scaling

Repeated adult (18+) enrolment spikes indicate migration inflow regions.

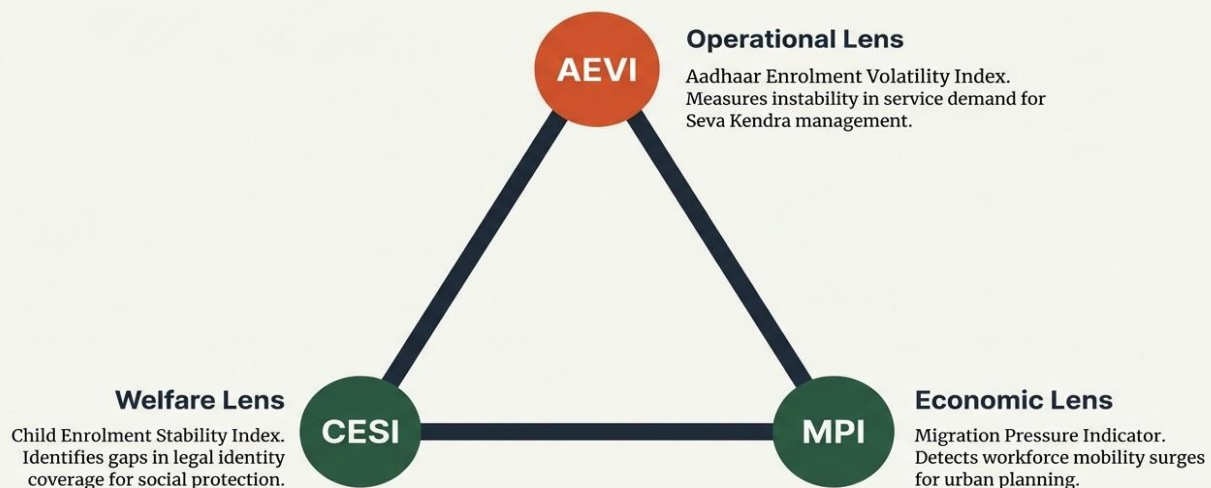
CESI — Child Enrolment Stability Index

Purpose: Assess stability of child enrolment (0–5, 5–17)

Policy use: Welfare outreach and accessibility planning

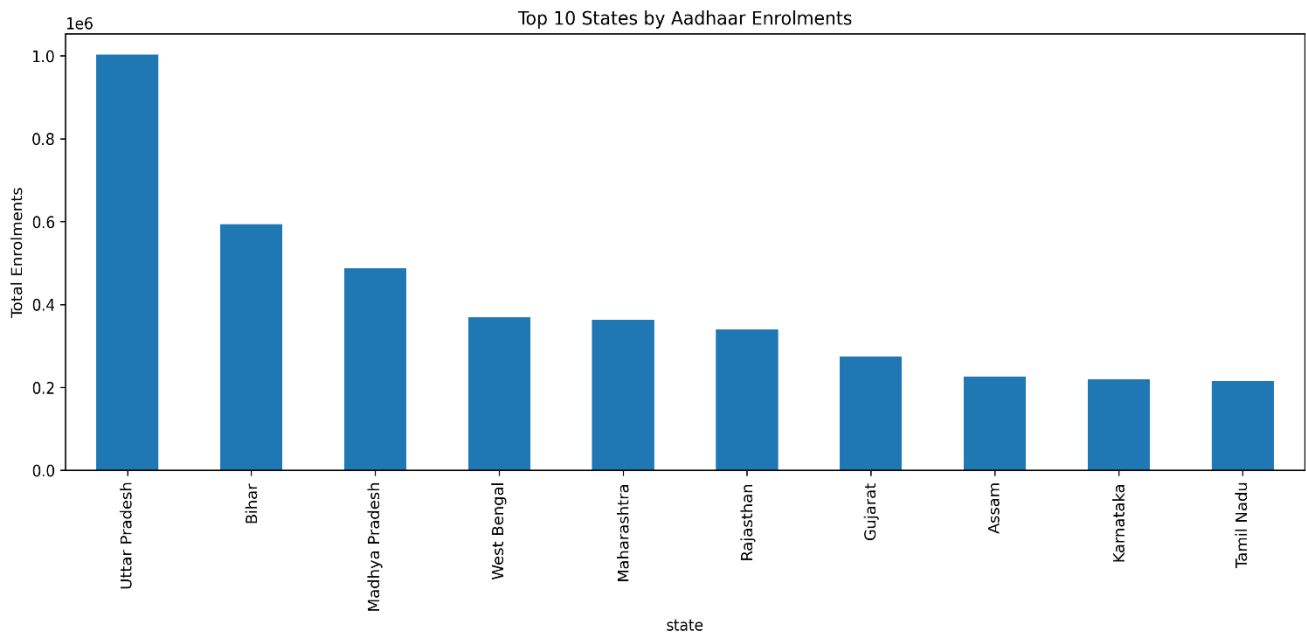
Low CESI values suggest gaps in child enrolment coverage

Three Lenses for Evidence-Based Governance



VISUALISATIONS

📍 STATE-WISE ENROLMENT DISTRIBUTION



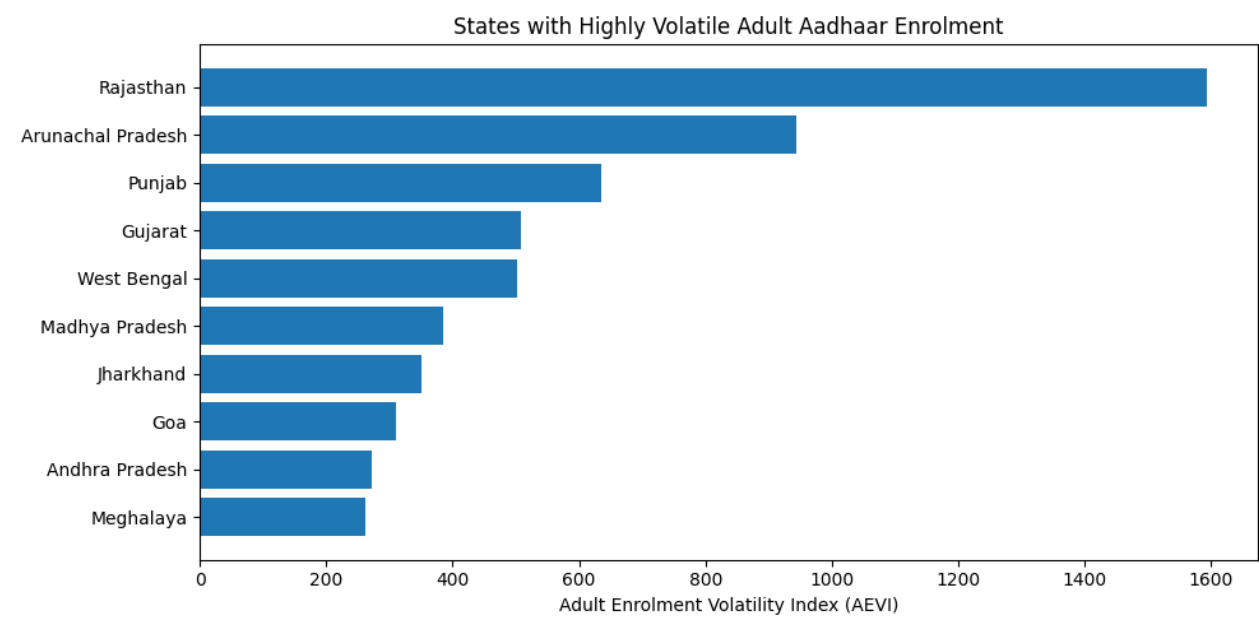
This graph shows:

- Total enrolment aggregated **state-wise**
- Bar chart with **average reference line**

Insight:

The state-wise enrolment distribution reveals significant concentration in a limited number of states, creating operational pressure points. States above the national average require higher staffing and infrastructure support.

📍 AEVI (AADHAAR ENROLMENT VOLATILITY INDEX)

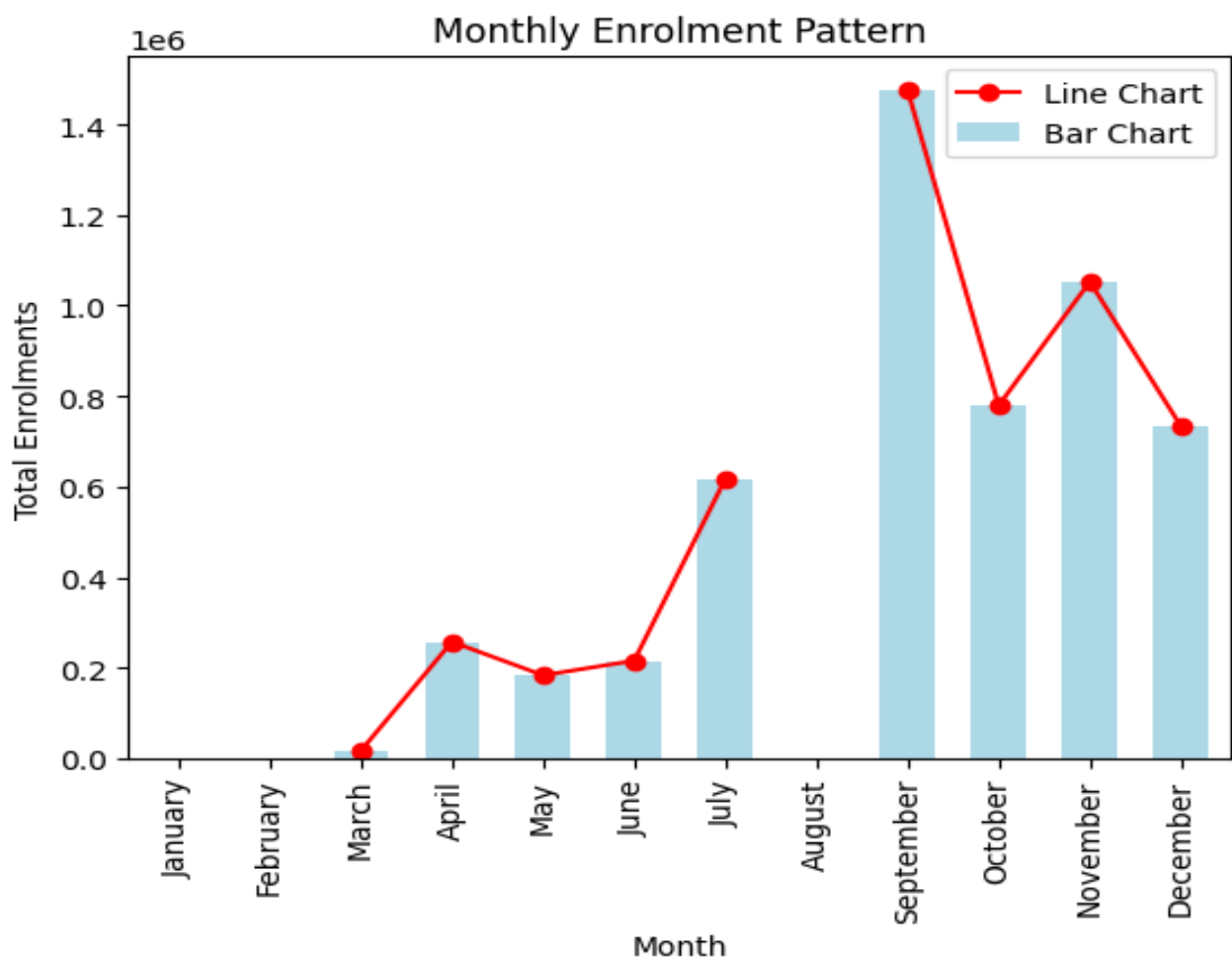


This graph shows:

- Computation of **AEVI per state**
- Based on **monthly enrolment fluctuations**
- Measures **irregularity over time**

Insight: The Aadhaar Enrolment Volatility Index (AEVI) quantifies temporal instability in enrolment demand. States with higher AEVI values experience irregular service demand, often linked to migration, awareness campaigns, or operational inefficiencies.

 Time-Series Trend of Total Enrolments



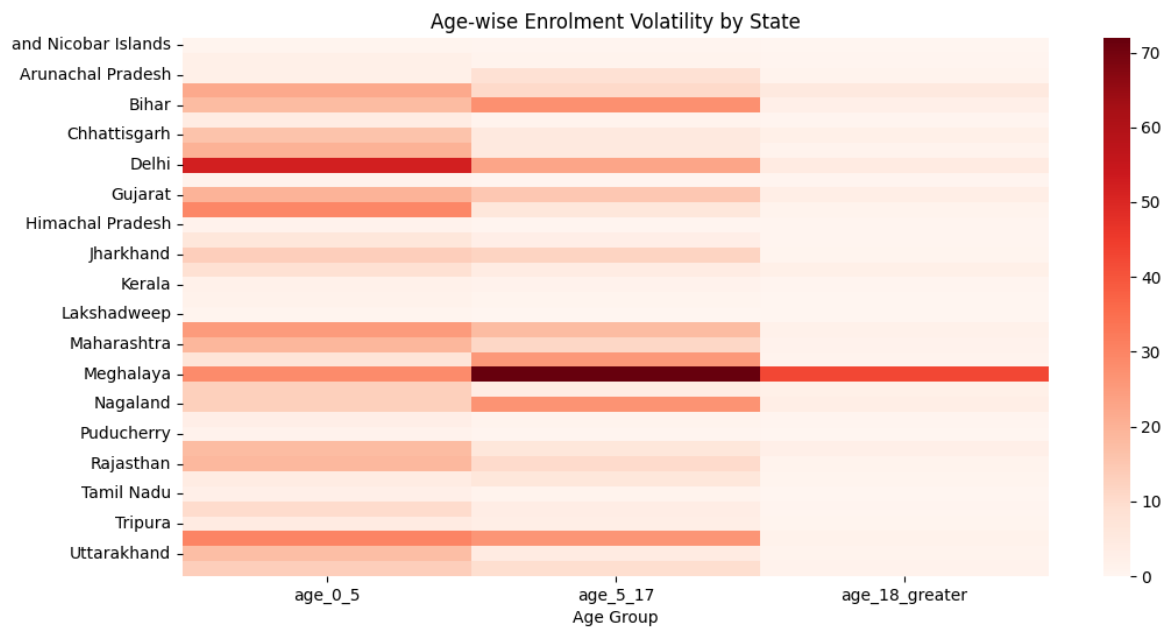
This graph shows: Shows demand evolution over time

Highlights volatility and seasonal demand patterns

Insight:

The time-series trend reveals periods of enrolment surges and stabilisation, enabling UIDAI to distinguish between structural demand growth and temporary spikes driven by campaigns or migration.

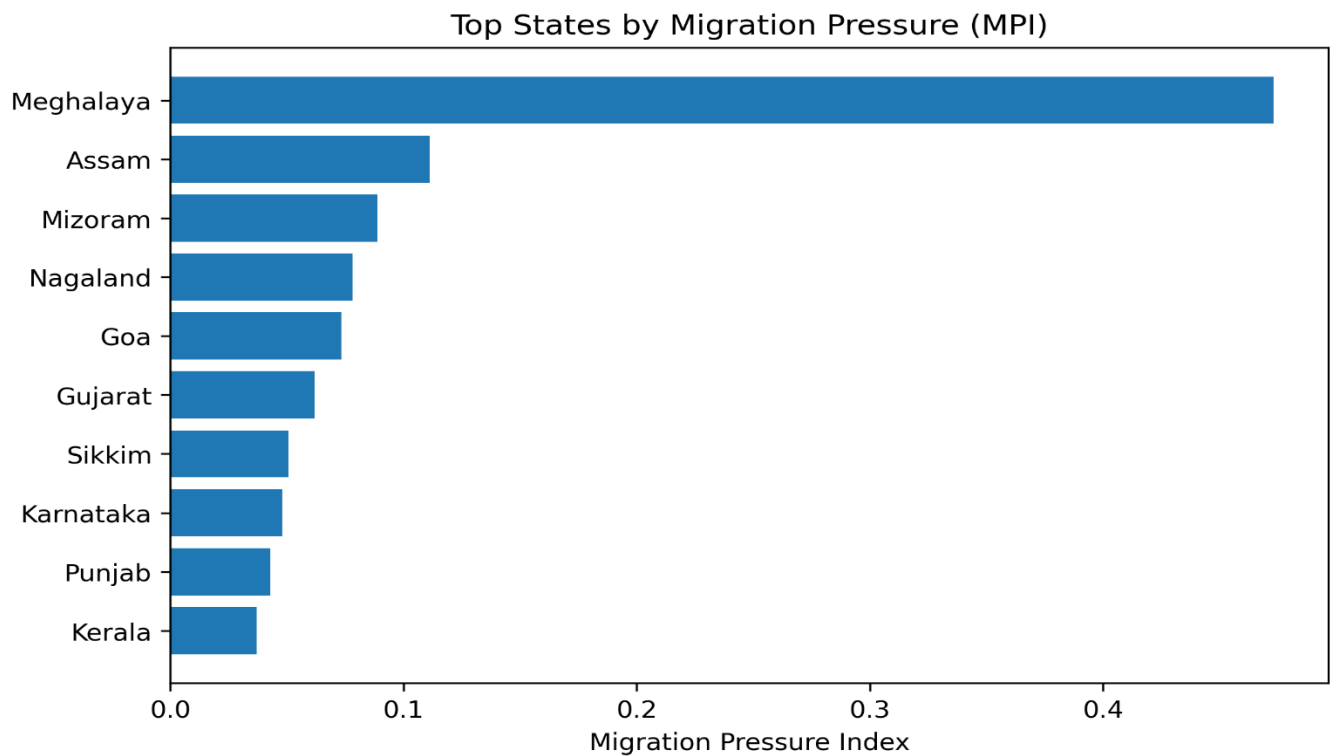
📍 Age-wise Enrolment Volatility Heatmap



- This graph shows:**
- Heatmap of **age-group-wise enrolment volatility**
 - Compares enrolment instability across:
 - States
 - Age groups (0–5, 5–17, 18+)

Insight: This heatmap highlights age-group-specific enrolment volatility across states. Higher instability in child age groups indicates potential gaps in awareness, accessibility, or administrative outreach, while stable patterns reflect mature enrolment ecosystems.

📍 Migration Pressure Indicator (MPI) — Adult (18+) Enrolment Spikes



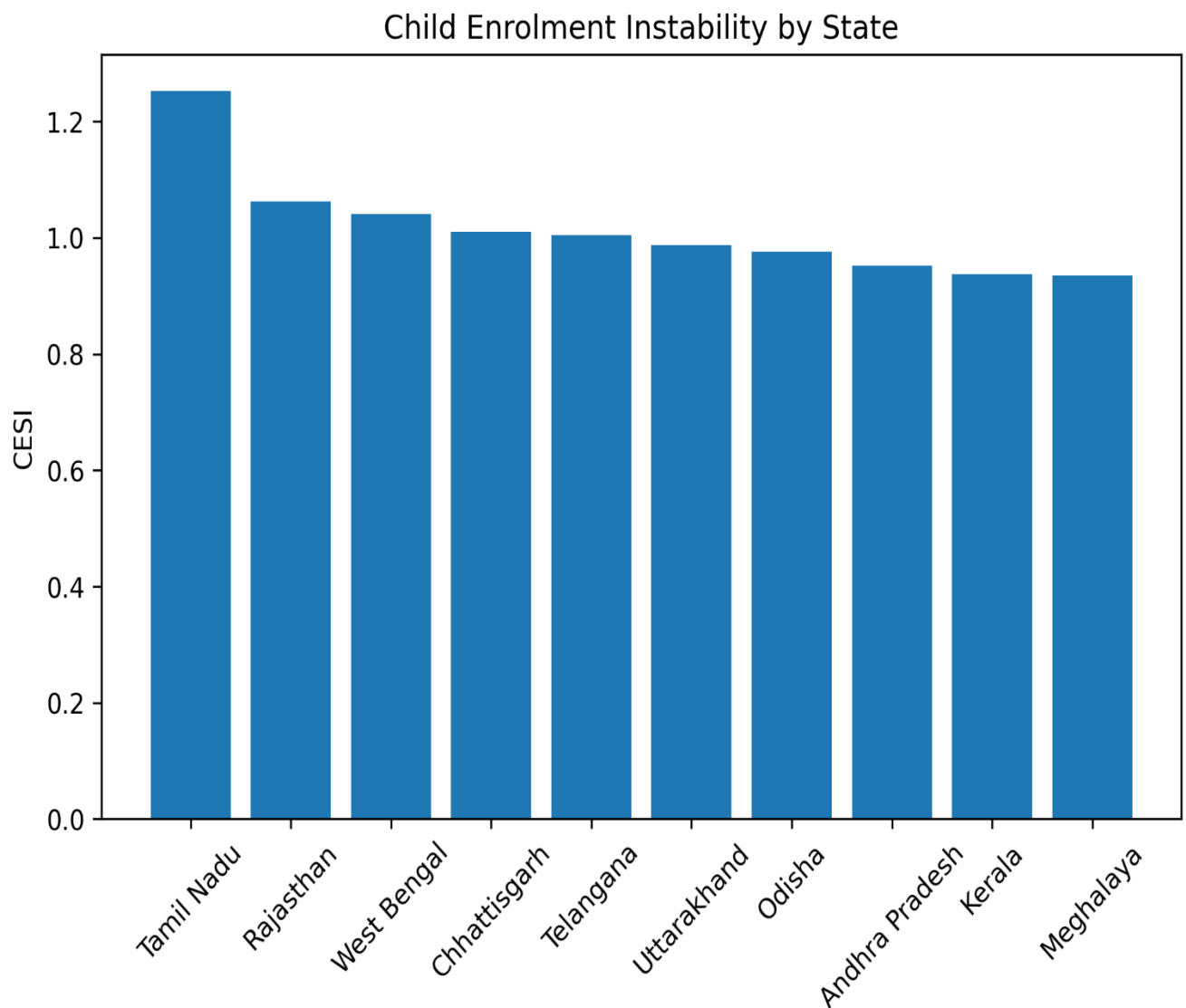
This Graph shows: Migration + governance relevance

During seasonal migration months, MPI spikes can trigger mobile enrolment units.

Insight:

Repeated spikes in adult enrolment indicate migration-linked service demand. States exhibiting consistent adult enrolment surges can be flagged as migration inflow regions requiring adaptive infrastructure.

✧ *Child Enrolment Stability Index (CESI) by State*



This graph shows: Welfare + inclusion focus

Insight:

CESI highlights disparities in child Aadhaar enrolment stability. Lower CESI values suggest regions where child coverage may be affected by accessibility, awareness, or administrative reach.

📌 *Enrolment Volatility Ranking (Top & Bottom States)*

state	Total_Adult_Enrolments	AEVI	Impact_Score	Risk_Impact_Index
Meghalaya	35078	262.099280	1.000000	262.099280
Rajasthan	5483	1594.278003	0.156309	249.199678
Gujarat	16063	507.171469	0.457922	232.245148
Assam	22555	204.332939	0.642996	131.385183
West Bengal	8495	501.166190	0.242175	121.369713
Madhya Pradesh	9476	385.254879	0.270141	104.073072
Punjab	3117	634.432890	0.088859	56.375145
Maharashtra	8103	225.421645	0.230999	52.072284
Uttar Pradesh	17699	96.326199	0.504561	48.602468
Karnataka	10038	147.598367	0.286162	42.237083

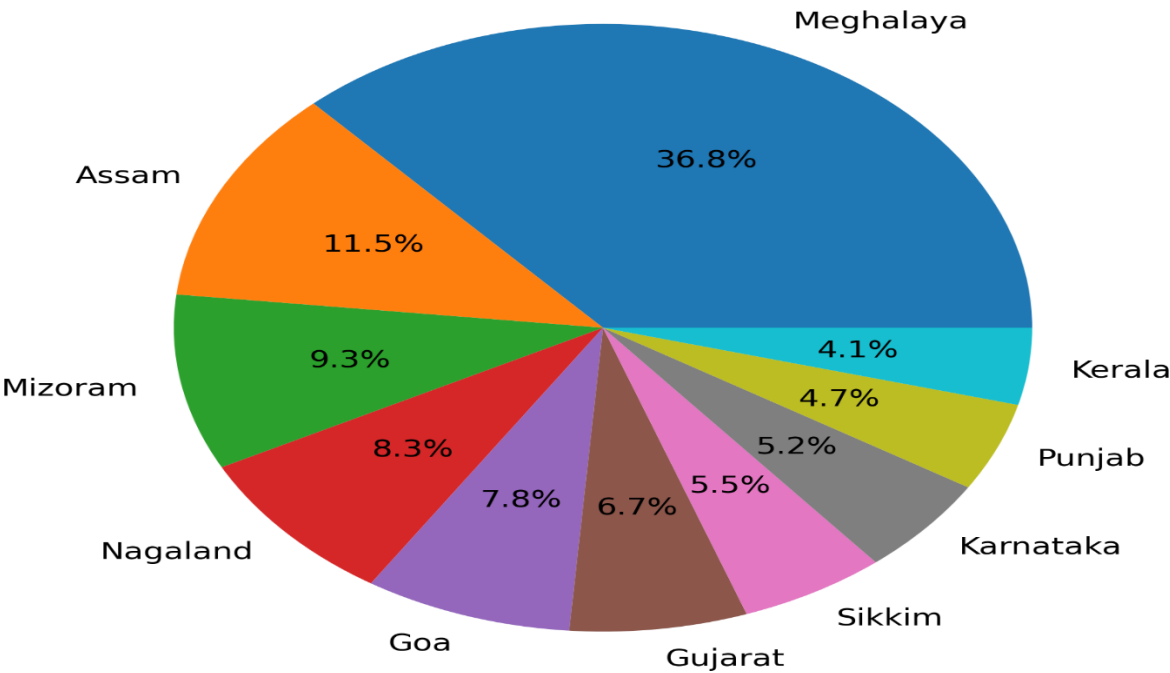
This Graph shows: Clear decision support

Insight:

Ranking states by enrolment volatility enables UIDAI to distinguish stable regions from high-risk regions, supporting proactive staffing and contingency planning.

📌 **Age-Group Contribution to Total Enrolments EDI**

States with Highest Aadhaar Service Dependency



Insight:
Age-group contribution analysis shows demographic demand structure. Dominance of adult enrolments in certain states reinforces migration-driven service pressure, while balanced distributions reflect stable enrolment ecosystems.

SOURCE CODE & REPRODUCIBILITY

- Entire analysis implemented in **Python**
- Libraries used: Pandas, NumPy, Matplotlib, Seaborn
- Executed via **Google Colab**
- Fully reproducible pipeline
- Clean, modular code blocks

[Colab Notebook](#)

IMPACT AND APPLICABILITY

This project demonstrates how anonymised Aadhaar enrolment data can be transformed into policy-relevant indicators without exposing individual-level information.

Applications include:

- Data-driven staffing decisions
 - Proactive infrastructure planning
 - Improved accessibility for migrant and vulnerable populations
- The proposed indices support scalable and privacy-preserving governance analytics.

LIMITATIONS AND CONCLUSION

Limitations

- Analysis is limited to aggregated enrolment data
- District-level deep dives can enhance localisation

Future Scope

- Integration with demographic and update datasets
- Improved predictive accuracy for service demand

Conclusion

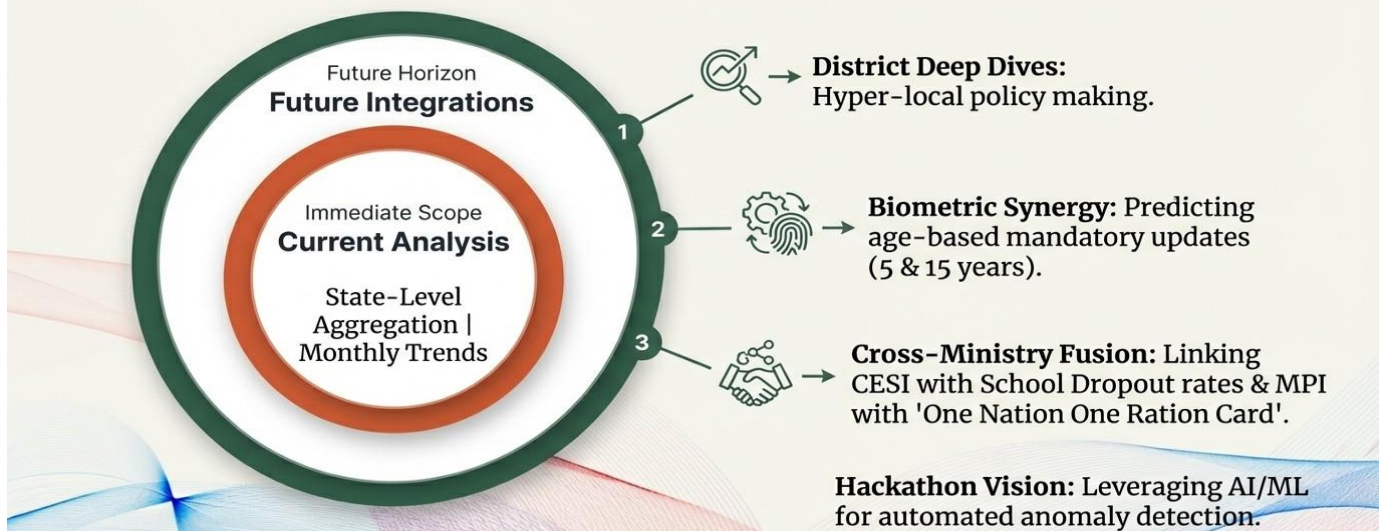
By introducing governance-oriented indicators beyond descriptive statistics, this project aligns Aadhaar data analysis with UIDAI’s mandate of efficiency, inclusion, and transparency.

From Reactive Administration to Proactive Governance

Dimension	Current State (Reactive)	Future State (Proactive)
Staffing	 Fixed Rosters	 Dynamic Resource Allocation (AEVI)
Demographics	 Static Records	 Migration-Responsive Planning (MPI)
Welfare	 Siloed Data	 Interoperable Protection Insights (CESI)

Impact: Reduced wait times, improved migrant support, and higher child protection coverage.

Future Scope & Integration



Unique Identification Authority of India
Government of India

Towards a Data-Responsive State

Aadhaar is more than an ID; it is a dynamic registry of the nation's pulse. By adopting AEVI, MPI, and CESI, UIDAI moves beyond managing identity to empowering society.



Recommendation: Embed these indicators into live administrative dashboards to drive efficiency, inclusion, and transparency.

